Failure Analysis Report

FAIL-0430 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0430
Failure Type	Synchronization Failure
Severity	Trivial
Component	Authentication
Application	APP-007 — InternetBrowser
Build Version	6.3.5
Timestamp	2023-05-23 02:07:53
Environment	ENV-001 — Mobile Lab
OS Version	Android 14
Network	WiFi-6
Resolution	Open

2. Test Case Reference

Test Case ID	TC-1594
Title	Verify that configuration reload fails gracefully with a user-friendly error whe...
Execution ID	EXEC-02090
Test Run	RUN-007 — Regression Cycle 7 — Build 5.3.1
Requirement	REQ-0109 — Authentication — Haptic Feedback

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete AR scene rendering within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the server returns a 503 Service Unavailable respo...

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-9346'; last_modified timestamps diverge by 83s.
```

Stack Trace / Log Excerpt:

```
com.samsung.smarthome.PushService.execute(line 715)
```

Log Reference	LOG-96613

5. Executed Steps at Time of Failure

SYNTHETIC DATA — No real Samsung confidential information

1	Initiate data synchronization by pressing the 'Search' button.	The audit log records the event with correct metadata.
2	Navigate to Settings > Bluetooth and tap 'Verify Identity'.	The device reconnects automatically within 10 seconds.
3	Capture a screenshot and attach as evidence.	The fallback UI is displayed with a human-readable error message.
4	Enter valid credentials: username='testuser_041', password='[REDA...	The audit log records the event with correct metadata.
5	Record response time using the built-in performance monitor.	The device reconnects automatically within 10 seconds.
6	Assert that database record for entity 'ENT-327' has status='SYNC...	The audit log records the event with correct metadata.

6. Root Cause Analysis

Health sensor sampling rate exceeds Bluetooth LE characteristic notification limit of 250ms intervals.

8. Historical Similar Failures

HIST-0192	FAIL-0193	Synchronization Failure	5.1.5	2022-02-02	6	Resolved
HIST-0266	FAIL-0267	Synchronization Failure	4.4.9	2022-03-21	10	Resolved
HIST-0429	FAIL-0430	Synchronization Failure	3.1.8	2023-11-07	10	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information